

Instructions: There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Let $f(x) = x^3 - 3x + 1$. First find all critical numbers of f , and then find the absolute maximum and absolute minimum values of f on the interval $[0, 3]$.

Solution: $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$
 $= 3(x+1)(x-1)$

$$f'(x) = 0 \Rightarrow 3(x+1)(x-1) = 0$$

$$x = -1, 1$$

Critical points: $-1, 1$.

end points: $0, 3$

$$f(-1) = (-1)^3 - 3(-1) + 1 = 3$$

$$f(1) = 1^3 - 3 \cdot 1 + 1 = -1 \rightarrow \text{absolute minimum value } f(1) = -1$$

$$f(0) = 0^3 - 3 \cdot 0 + 1 = 1$$

$$f(3) = 3^3 - 3 \cdot 3 + 1 = 19 \rightarrow \text{absolute maximum value } f(3) = 19$$

Problem 2. (4 points) If $f(1) = 5$ and $f'(x) \leq 3$ for all x , how large can $f(5)$ possibly be? (using the Mean Value Theorem.)

Solution: By the Mean Value Theorem, we have.

$$f'(c) = \frac{f(5) - f(1)}{5 - 1} \quad \text{for some } c \text{ in } (1, 5).$$

$$\Rightarrow f(5) = f(1) + f'(c)(5 - 1)$$

$$= f(1) + 4f'(c)$$

$$\leq 5 + 4 \cdot 3 = \underline{17}$$